

ORIGINAL ARTICLE

Effect of conspecific and heterospecific urine odors on the foraging behavior of the golden spiny mouse

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Abstract

The common spiny mouse (*Acomys cahirinus*) inhibits the foraging activity of the golden spiny mouse (*A. russatus*). These two sympatric species of spiny mice, which are considered habitat competitors, occur in extreme arid environments. To test this theory of competition, the influence of urinary odors of both conspecific and heterospecific mice on the foraging behavior of *A. russatus* was studied under controlled laboratory conditions. Twenty adult males, born in captivity and unfamiliar to the odors of the donor mice, were tested in 3 experimental conditions choosing between 2 seed patches that were scented with urine of either heterospecifics (*A. cahirinus*), conspecifics (*A. russatus*) or controls (odors of the tested individual). Of the 20 males, 12 were also tested with urine of unfamiliar gerbils, bushy-tailed jird, considered as competitors in the field. Both conspecific and heterospecific urine samples from *Acomys* significantly reduced foraging behavior of *A. russatus* when compared to the control odor. The inhibitory effect of the *Acomys* urine does not result from the novelty of chemical stimuli of the urine because no effect was shown with unfamiliar gerbil urine. The findings are in accordance with the general theory that *A. cahirinus* dominates the foraging activity of *A. russatus*. We hypothesize that chemical cues in the urine of *Acomys* spp. might induce a negative effect on the foraging behavior of *A. russatus*.

Key words: *Acomys* spp., foraging behavior, golden spiny mouse, *Sekeetamys calurus*, urine odors

INTRODUCTION

Several species may coexist in the same habitat as a result of their respective adaptation to particular environmental conditions (Gause 1934). Specific adapta-

tions include choice of habitat and microhabitat, as well as other behavioral and physiological traits (e.g. activity rhythms, shelter, diet and foraging behavior). In some cases, there is no total separation between ecological niches and, therefore, competition for resources may occur. Such a competitive process has been described in 2 species of spiny mice of the genus *Acomys*, the common spiny mouse (*Acomys cahirinus* Desmarest, 1819) and the golden spiny mouse (*Acomys russatus* Wagner, 1840), which are sympatric in the rocky habitats of the most arid parts of Israel's Rift Valley (Shkolnik 1971; Harrison & Bates 1991; Shargal *et al.* 2000; Kronfeld *et*

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al. 1994) and compete for food (Kronfeld-Schor & Dayan 1999; Jones *et al.* 2001) and shelter (Haim & Rozenfeld 1995; Shargal *et al.* 2000). The coexistence of these 2 species is facilitated by activity displacement of *A. russatus* from nocturnal to diurnal habits, with harsher conditions for living due to the higher diurnal temperature and solar radiation. Removal of *A. cahirinus* from the common habitat allows *A. russatus* to return to a nocturnal activity (Shkolnik 1971). The temporal partitioning between spiny mouse species (Gutman & Dayan 2005) is probably not due to aggressive interference between them (Pinter-Wollman *et al.* 2006). Among several potential proximal mechanisms, chemical cues (urine and feces) released by *A. cahirinus* mediate a phase shift in daily rhythms of body temperature, activity and metabolic rates of nocturnal *A. russatus* (Fluxman & Haim 1993; Haim & Rozenfeld 1993; Haim & Fluxman 1996; Friedman *et al.* 1997). However, almost no information exists concerning the scent marking activity of *A. cahirinus*, although scent marks probably play a role in a specific social context (Brown & MacDonald 1985). Marking activity has been previously described for *A. russatus* in a social context (Rozenfeld *et al.* 1994). Other studies reveal the importance of olfactory cues during the development of *A. cahirinus* (Porter *et al.* 1980) and for food preferences (Porter *et al.* 1990). Chemical cues might mediate the inter-specific relationships between these mice, as described in other rodent species and subspecies (de Jonge 1980; Todrank & Heth 1996; Christophe & Baudoin 1998). For example, it has been demonstrated in several mammalian species that odor of a competitor can be as effective as the competitor itself for inducing behavioral responses (Barnard & Brown 1985; Devenport *et al.* 1999; Jacquot & Baudoin 2002). The results of a field experiment under natural conditions by Dobly *et al.* (2001) showed that fresh semiochemical signals (in the urine and feces) released by *A. cahirinus* decreased the number of *A. russatus* that foraged and touched the food, while it increased the rate of sniffing from a distance. In this field experiment the authors did not compare *A. russatus* behavior in response to conspecific odors because the results of previous studies have shown that, in regards to activity and body temperature daily rhythms, only heterospecific odors have an effect on *A. russatus* (Haim & Rozenfeld 1993). Taking into account the competition between *A. cahirinus* and *A. russatus* under natural conditions and the dominance of *A. cahirinus*, the aims of the present study are: (i) to characterize the effect of odors of *A. russatus* on *A. russatus* individuals' foraging behavior; (ii) to

verify, in a controlled environment and with a higher homogeneous sample of mice than in Dobly *et al.* (2001), the conclusion of an inhibitory effect of *A. cahirinus* odors on the foraging behavior of *A. russatus*; and (iii) to characterize the effect of odors of another competitor rodent species (Haim & Rozenfeld 1998), the bushy-tailed gerbil (*Sekeetamys calurus* Thomas, 1892) on the foraging behavior of *A. russatus*. This gerbil species has the same distributional pattern as *A. russatus*. In places where *A. cahirinus* does not exist at high altitudes, the former is nocturnal while *A. russatus* is diurnal, indicating possible competition between mice and gerbils (Haim & Rozenfeld 1998). In this manuscript, we use the name *A. cahirinus* for the common spiny mouse instead of *A. dimidiatus* (Barome *et al.* 2001). We prefer to use *A. cahirinus*, which is suggested by Bates (1994) and this name adopted in most recent papers published on the common spiny mice from Israel.

MATERIALS AND METHODS

Test animals

A total of 20 laboratory born adult males *A. russatus* were used in the present study (mean body mass $\pm$ SD = 66.5 ± 10 g). Individuals were weaned at 12 days and then kept in groups of 3 individuals of the same sex in standard polycarbonate cages. During experiments, mice were kept in a controlled room with an ambient temperature of 28 ± 3 °C under a photoperiod of 16:8 h L:D, with light on at 06.00 hours. Rat pellets were provided *ad libitum*; carrots and cucumbers were supplied as a water source and were given in the morning once a day. To homogenize their social status (deCatanzaro & Gorzalka 1979; Féron & Baudoin 1998), mice were isolated for 48 h in standard cages prior to being introduced into the experimental arena. The tested individuals were not exposed to the odors of other species prior to the tests as they were bred in separate rooms in the breeding colony.

Methods

Odor stimuli

Pooled urine samples were collected from groups of 3 males of *A. russatus*, *A. cahirinus* and *S. calurus* using a non-invasive procedure. Each group was socially stable without any particular perturbation in their environment. Urine of conspecifics was collected from unfamiliar and not closely related males from the same breeding colony. Urine samples were collected in a plastic tray beneath the cages in the early morning and were trans-

ferred to Eppendorf tubes using glass Pasteur pipettes. Urine samples were then kept for a few hours in the refrigerator (4 ± 1 °C) until testing, as we know that only fresh chemical signals have a behavioral effect, as observed in field conditions (Dobly *et al.* 2001).

Protocol and experimental method

Our study was organized in 3 phases.

Phase 1 (habituation period). Each male was allowed to mark by using traces of urine, feces and body odorants such as oral secretions (Rozenfeld *et al.* 1994) in the opaque plastic arena ($70 \times 50 \times 40$ cm) with 2 empty Petri dishes (10 cm diameter) for 24 to 48 h. Food (carrots, rat pellets and a limited quantity of seeds) was available during this period. At the end of this phase, the 2 Petri dishes were removed and kept in a plastic bag in the refrigerator to be used as 'control' odors in the second phase.

Phase 2 (experimental period). In the morning before the first test, we removed food items from the arena. Then, for each 30 min test, we introduced simultaneously 2 Petri dishes, each with 40 seeds with 20 g of clean sand over the seeds. As a source of odor, 5 drops of urine were added in a regular pattern over the sand and seeds in each patch. The Petri dishes were introduced into the arena approximately 60 cm from the resting subject and 15 cm away from the median separation in one extremity of the arena. Mice did not show fear or freeze during this introduction and they generally started to explore the arena as soon as the dishes were introduced. In the 'control' condition, mice were tested for responses to their own odors collected during the habituation period, with the same quantity of sand and seeds as in the experimental dishes and with 5 drops of distilled water over the sand. Mice had the same type of visual and tactile information concerning the food patches (dishes with sand + drops and seeds) both in the experimental and in the control situations. Then, each mouse was tested with conspecific or heterospecific urine in a two-choice test. During the 3 days after Phase 1, tests were conducted with the order and the side of presentation of odors determined by chance: odors of *A. cahirinus*/control; *A. russatus*/control; *A. cahirinus*/*A. russatus*. Of the 20 adult *A. russatus* males, 12 were also tested during these 3 days with seed patches scented with urine odors of *S. calurus* (Haim & Rozenfeld 1998) with the same protocol. At the end of each experimental day, all Petri dishes were removed and food (carrots, rat pellets and a limited quantity of seeds) was distributed and available during the night.

Phase 3 (laterality and satiety tests). After these 3 experimental days, to test a possible laterality (left and right Petri dishes) effect or a satiety (the number of seeds eaten in first visited dish and in the other dish) effect on food consumption, each mouse was tested 3 times with 2 dishes scented only with their own odors ('control' dishes), with a minimum of 2 h between trials. For each of these 3 trials, we noted the number of seeds eaten. After the period of observation, mice were returned into their previous cages with same-sex familiar congeners, without inducing any aggression. Arenas were cleaned after each 5 to 6 day period (habituation + tests) with water and ethanol, before being used with other tested mice.

Behavioral data

Mice behavior was recorded directly by observation on a standard grid with time and activities in the arena (order of choice of the dishes, feeding in the patches, locomotion, grooming, freezing or resting). The number of seeds remaining in dishes was counted after the testing procedure without any information given about odor conditions.

Ethical note and legislation

Animal care and experimental procedures were conducted in respect of legislation and with the agreement of the University of Haifa Ethical Committee. The authorization to experiment (93-031) was delivered to CB by the 'Préfecture de Seine-Saint-Denis' (France).

Statistical analyses

We used the permutation test (also called the randomization test) to test the comparison of means for paired samples and the Pearson correlations between numerical variables. This non-parametric test does not need a rank classification and maintains all the available information (Edgington & Onghena 2007; Good 2005). There is no condition for normality of distribution of the parent populations. The laterality and satiety effects were studied using the permutation test for paired samples applying the exact method (StatXact-8 software, Cytel, 2007). For each experimental test, the means of the presence duration in the 2 dishes were compared using the same statistical test, which was also used for comparing the means of seeds eaten in the 2 dishes. Moreover, for each experimental situation, the distribution of the number of mice present in the 2 dishes, associated to the first chosen dish, was compared to a uniform distribution using a binomial test (StatXact-8 software, Cytel, 2007).

RESULTS

The 3 trials performed with 2 control dishes (Phase 3) showed no significant effect of laterality or satiety. There was no significant laterality effect on the side of the first dish that was sniffed: for the right side, first trial = 7/20, second trial = 10/20 and third trial = 13/20. These proportions did not differ significantly from 50% (respectively, $P = 0.26$, $P = 1$, $P = 0.26$). There was also no significant laterality effect on the number of seeds eaten by the mice. On the left side (mean $\pm$ SD), 34.20 ± 3.93 seeds were eaten, and on the right side, 34.15 ± 4.49 seeds ($P = 0.97$). We also did not observe any satiety effect on the number of seeds eaten between the first and the second dishes: on the totality of the 3 trials, the number of seeds eaten in the first dish and in the second dish were similar, respectively (mean $\pm$ SD), 34.28 ± 4.06 seeds and 34.07 ± 3.99 seeds, and did not

differ significantly ($P = 0.75$). The experimental trials showed a negative effect of the urine of *Acomys* spp. on the foraging behavior of *A. russatus* compared with the controls.

Figure 1 shows that the mean number of seeds eaten was significantly reduced, compared with the control, both in dishes with odors of *A. cahirinus* ($P < 0.01$) or *A. russatus* ($P \leq 0.005$), without significant difference between them ($P = 0.23$). Figure 2 shows that mice spent less time foraging in dishes with odors of *Acomys* spp. (*A. cahirinus*, $P = 0.003$; *A. russatus*, $P = 0.005$) than in control dishes, without a significant difference between these 2 types of dishes with *Acomys* spp. urine ($P = 0.73$).

We observed a strong positive correlation between the number of seeds eaten and the time spent in the patches perfumed with *Acomys* odors (in different con-

Figure 1 Golden spiny mice *Acomys russatus* foraging behavior in a two-choice test between scented dishes: number of seeds eaten (mean $\pm$ SE) in 3 situations (Ac = odors of *A. cahirinus*, Ar = odors of *A. russatus*, C = control or own odors of the tested individual). ** $P < 0.01$

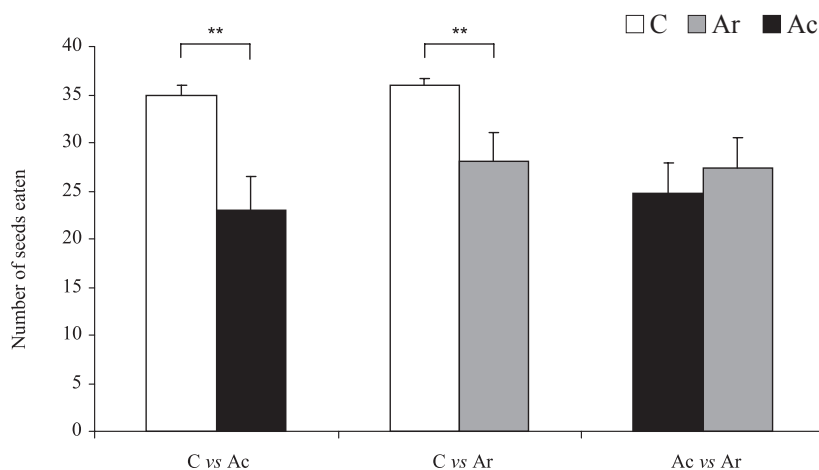
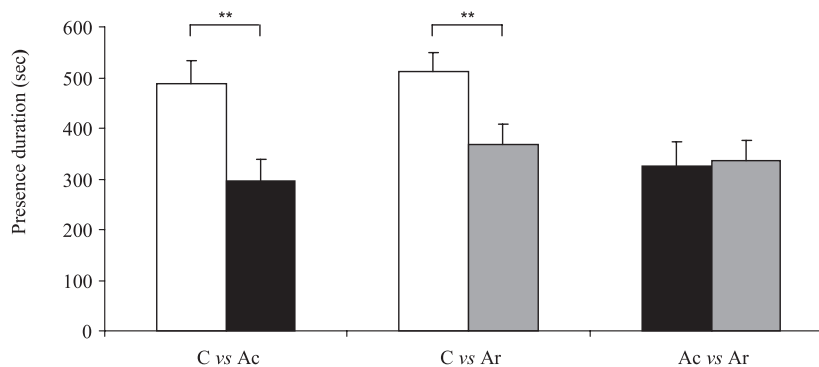


Figure 2 Golden spiny mice *Acomys russatus* foraging behavior: time spent in patches with seeds (mean $\pm$ SE) in 3 situations (Ac = odors of *A. cahirinus*, Ar = odors of *A. russatus*, C = control or own odors of the tested individual). ** $P < 0.01$



texts): patches with odors of *A. cahirinus* (in the tests with control), $r = 0.94$, $P < 0.001$; odors of *A. cahirinus* (in the tests with *A. russatus* odors), $r = 0.88$, $P < 0.001$; *A. russatus* (tests with control odors), $r = 0.81$, $P < 0.001$; *A. russatus* (tests with *A. cahirinus* odors), $r = 0.92$, $P < 0.001$. We did not observe similar correlations between these 2 dependant variables (seeds and time) when considering the patches with the mice's own odors (controls): control odors (in the tests with *A. cahirinus* odors), $r = 0.35$, $P = 0.12$; control odors (tests with *A. russatus*), $r = -0.05$, $P = 0.83$. In exposure to *S. calurus* urine (tests with control odors), the number of seeds eaten in the 2 dishes (control odors vs *Sekeetamys* odors: 36.25 ± 0.81 seeds vs 34.25 ± 1.40 seeds) did not differ significantly ($P = 0.10$), nor did the time spent in the 2 patches (control vs *Sekeetamys* odors: 453.2 ± 44.1 sec vs 436.7 ± 26.8 sec, $P = 0.74$). Familiar own odors (controls) of *A. russatus* mice and unfamiliar odors of *S. calurus* induced no change in mice foraging behavior.

DISCUSSION

Our hypothesis that there is an inhibitory effect of the odor of *A. cahirinus* on *A. russatus* behavior was confirmed as we observed a reduction in the time spent in the food patches and in the number of seeds eaten when *A. cahirinus* urine odors scented the patch compared with the controls. When mice had a choice between patches perfumed with odors of the 2 *Acomys* species, the quantity of seeds eaten and the time spent in the patch with *A. cahirinus* odors were very similar to those recorded in the first combination (*A. cahirinus* vs control). In these 2 tests, there was a strong positive correlation between the number of seeds eaten and the time spent in the patches with the heterospecific (*A. cahirinus*) odors. Results of a previous study revealed that fresh urine and feces of *A. cahirinus* reduced the number of *A. russatus* individuals, observed per session of observation, compared with sessions without *A. cahirinus* odors, considered as controls by Dobly *et al.* (2001). In that experiment, only a few individuals of the 2 sexes (3 males and 5 females) were studied. Our study, with controls and with more individuals (20 mice) and a homogenous (only males) subject sample, confirms and details the inhibitory effect of unfamiliar *A. cahirinus* odors on *A. russatus* foraging behavior. Moreover, our results with laboratory born mice seem to indicate that, at least in the conditions of our study, this effect could be independent of the individual experience because these tested mice had not come in contact with *A. cahirinus* individuals before the testing period. It is not

easy to interpret this result because in some other species of rodents, individuals need prior experience with heterospecific individuals to respond to the presentation of their odors (Gouat *et al.* 1998; Krasnov & Khokhlova 1996). In another context, an innate response of prey to the odors of predators is well documented in several species of mammals (in rats [Cattarelli 1982] and in sheep [Arnould & Signoret 1993; Arnould *et al.* 1998]), but it is unlikely that the coexistence under natural conditions between these 2 competing *Acomys* species, during a relatively short period of time, favored the development of a similar mechanism. Most probably, under natural conditions, mice of these 2 sympatric species have the opportunity to encounter mice of the other species or their odors in particular sites (like shelters, and feeding and nesting places) and to become familiarized with them. If mice of the 2 species are resident in a particular site, we could expect that the individual experience would play an important role, as has been described in other species, with the possible consequence of reducing negative effects, similarly to the 'dear enemy' effect described by Fisher (1954) or, in contrast, increasing the reactions to the presence of a close competitor individual. In what follows, the inhibitory response of laboratory born *A. russatus* to the odors of *A. cahirinus* will be discussed further in regards to our other results.

The second main result observed in our study is an inhibitory effect of conspecific odors on *A. russatus* foraging behavior, very similar to the effect observed with *A. cahirinus* odors. The tested mice were not familiar with the conspecific odors used during testing. It is possible that this lack of familiarity with the experimental stimuli is responsible for this inhibitory effect. It would be interesting to observe comparatively the effect of familiar odor of conspecifics (e.g. kin individuals or neighbors, etc.) on foraging activity. We know that in a foraging context, *A. russatus* are very sensitive to odors present on the food sources. It has been shown that when food is marked with oral scent, males spend less time sniffing sunflower seeds that they had previously handled than seeds handled by unfamiliar males (Rozenfeld *et al.* 1994). Urine odors might also signal that a site has already been foraged and that it would be more beneficial to explore a new site than to spend time in the area with urine marks, as has been reported in ground squirrels *Spermophilus spilosoma* Bennett, 1833, *S. mexicanus* Wagner, 1843 and *S. tridecemlineatus* Audubon and Bachman, 1849 (Livoreil 1994; Livoreil & Baudoin 1996). We also expect that the same process, as in

ground squirrels, would occur with the individual odors of a mouse when it explores food patches. In our experimental setup, the presence of the mouse's own odors in the food patches did not induce a similar effect on the foraging of *A. russatus*. If the lack of familiarity with the odors was the factor explaining the effect of odors of *Acomys* spp., we would expect a similar effect induced by the odors of the gerbil *S. calurus*, a putative competitor in field conditions (Haim & Rozenfeld 1998). This was not the case in our study and unfamiliar odors of the sympatric gerbil *S. calurus* in the food patches induced no effect on the foraging behavior of male *A. russatus*. Consequently, the lack of familiarity with odors is not sufficient for explaining the inhibitory effect on foraging behavior induced by *Acomys* urine odors. Therefore, it seems reasonable to consider now, as a provisory conclusion, that both the species-specific odors and familiarity may be 2 important co-acting factors.

The inhibitory effect of unfamiliar conspecifics odors might also be explained by urine compounds associated, by learning, with competitive situations during social interactions. To the best of our knowledge, there is no detailed study of the social organization of *A. russatus*, but we can infer from the low local population density described in field studies (Haim & Borut 1975; Shargal *et al.* 1998, 2000) that *A. russatus* are probably not highly social animals, even if an overlapping of male and female home ranges exists (Shargal *et al.* 2000). Under the laboratory conditions of our study, mice were never isolated, except before the test for a short time. They were living in small stable groups and fed *ad libitum*. Under these conditions, agonistic interactions were rare in stable groups. We cannot exclude the possibility, however, that laboratory born individuals might associate conspecific chemical cues with competitive or agonistic contexts.

Did we observe other differences between responses to the 2 types of urine odors (*A. cahirinus* and *A. russatus* odors)? During the tests we carried out, 4 out of the 20 *A. russatus* individuals froze in reaction to the presence of *A. cahirinus* urine odors and did not forage at all. They did not react this way in the presence of unfamiliar urine odors of conspecifics. Moreover, Haim and Rozenfeld (1998) report that, in enclosures, *A. russatus* avoided *S. calurus* by escaping or freezing. It would be interesting to study the freezing reaction and the emotional reactivity of *A. russatus* to various odor types and combinations. Jones and Dayan (2000) state, for example, that *A. russatus* mice showed more cautious responses to fecal odors of the fox *Vulpes cana* Blanford,

1877 than did *A. cahirinus*. Our preliminary result on the *A. russatus* freezing reaction needs to be complemented with more testing of individuals before definite conclusions can be drawn. At this stage, it is the only behavioral difference (not statistically significant) observed between the effects induced by the 2 types (conspecific and heterospecific) of *Acomys* spp. urine.

The results of our study are not contradictory to the 'competitor hypothesis', but a parsimonious hypothesis can be proposed from our study. This possibility invokes the role of common chemical cues in urine of these 2 species of *Acomys*. The inhibitory effect would then be similarly induced by common chemical compounds produced by individuals of both species. The lack of effect of their own odors might be due to their permanent presence inducing a physiological habituation. Further studies are also needed using comparative chemical analysis of urine of the 2 spiny mouse species.

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